

EXPERIMENT 13 – File Organization Techniques

Aim: To implement Sequential, Direct (Random), and Indexed File Organization.

Program A – Sequential File Organization

C Program:

```
#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

int main()
{
    FILE *fp;
    struct student s;

    fp = fopen("student.dat", "w");

    printf("Enter Register Number: "); scanf("%d", &s.regno);
    printf("Enter Name: "); scanf("%s", s.name);

    fprintf(fp, "%d %s\n", s.regno, s.name);
    fclose(fp);

    fp = fopen("student.dat", "r");
    fscanf(fp, "%d %s", &s.regno, s.name);

    printf("\nRecord Details\n");
    printf("Register Number : %d\n", s.regno);
    printf("Name : %s\n", s.name);
    fclose(fp);

    return 0;
}
```

Output (Input: RegNo=101, Name=Arun):

```
Record Details
Register Number : 101
Name : Arun
```

Program B – Direct (Random) File Organization

C Program:

```
#include <stdio.h>

struct student { int regno; char name[20]; };

int main()
{
    FILE *fp;
    struct student s;

    fp = fopen("random.dat", "wb+");

    printf("Enter Register Number: "); scanf("%d", &s.regno);
    printf("Enter Name: "); scanf("%s", s.name);

    fwrite(&s, sizeof(s), 1, fp);
    rewind(fp);
    fread(&s, sizeof(s), 1, fp);

    printf("\nRecord Found\n");
    printf("Reg No : %d\n", s.regno);
    printf("Name : %s\n", s.name);

    fclose(fp);
    return 0;
}
```

Output (Input: RegNo=102, Name=Kumar):

```
Record Found
```

Reg No : 102
Name : Kumar

Program C – Indexed File Organization

Shell Script:

```
#!/bin/bash
```

```
echo "Enter Student Records"
```

```
echo "101 Arun" > index.txt
```

```
echo "102 Kumar" >> index.txt
```

```
echo "103 Ravi" >> index.txt
```

```
echo "Enter Register Number to Search:"
```

```
read key
```

```
grep "^$key" index.txt
```

Result: Sequential, Direct, and Indexed File Organization techniques implemented successfully.
